

TRAMOX Capsule 50mg



Composition

Tramadol hydrochloride 50mg

Product Description

Buff, hard gelatine capsule, size 3, containing white to off-white granular powder.

Pharmacodynamics

Tramadol hydrochloride is a centrally acting analgesic. It is non selective pure agonist at mu, delta and kappa opioid receptors with a higher affinity for the mu receptor. Other mechanisms which may contribute to its analgesic effect are inhibitor of neuronal reuptake of noradrenaline and enhancement of serotonin release.

Pharmacokinetics

After oral administration, tramadol is almost completely absorbed. Mean absolute bioavailability is approximately 70% following a single dose and increases to approximately 90% at steady state. Plasma protein binding of tramadol is approximately 20%. When C14 labelled tramadol was administered to humans, approximately 90% was excreted via the kidneys with the remaining 10% appearing in the faeces. Tramadol has a linear pharmacokinetic profile within the therapeutic dosage range. The half life of the terminal elimination phase ($t_{1/2\beta}$) was 6.0 ± 1.5 h in young volunteers. Tramadol pharmacokinetics show little age dependence in volunteers up to the age of 75 years. In volunteers aged over 75 years, $t_{1/2\beta}$ was 7.0 ± 1.6 h on oral administration.

Following a single oral dose administration of tramadol 100mg as capsules or tablets to young healthy volunteers, plasma concentrations were detectable within approximately 15-45 minutes with a mean C_{max} of 280 to 308 mcg/L and T_{max} of 1.6 to 2h.

Tramadol is metabolized by the cytochrome P450 iso-enzyme CYP2D6. It undergoes biotransformation to a number of metabolites mainly by means of N- and O-demethylation. O-desmethyl tramadol appears to be the most pharmacologically active metabolite, showing analgesic activity in rodents. As humans excrete a higher percentage of unchanged tramadol than animals it is believed that the contribution made by this metabolite to analgesic activity is likely to be less in humans than animals. In humans the plasma concentration of this metabolite is about 25% that of unchanged tramadol.

Since tramadol is eliminated both metabolically and renally, the terminal half-life $t_{1/2\beta}$ may be prolonged in impaired hepatic or renal function. In patients with liver cirrhosis $t_{1/2\beta}$ tramadol was a mean of 13.3 ± 4.9 h; in patients with renal insufficiency (creatinine clearance < 5 ml/min) it was 11.0 ± 3.2 h.

Indication

Moderate to severe pain.

Recommended Dose

The dosage should be adjusted to the intensity of the pain and the sensitivity of the individual patient. Unless otherwise prescribed, Tramox Capsule should be taken as follows:

Single dose for adults and adolescents over 12 years of age: 1-2 capsules (50-100mg tramadol hydrochloride) to be taken with a little liquid. If pain relief is unsatisfactory, a further capsules may be taken after about 30-60 minutes. If in severe pain the analgesic demand is likely to higher, the higher dose (2 capsules) can be taken as initial dose. The lowest analgesically effective dose should generally be selected. Daily doses of 400mg tramadol hydrochloride should not be exceeded, except in special clinical circumstances. For the treatment of severe post-operative pain, on-demand pain relief may require even higher doses in the first few hours. In general, over 24 hours more than the normal doses are not necessary.

Tramox is not approved for use in patients below 12 years old.

Paediatric population

The safety and efficacy of Tramadol has not been studied in the paediatric population. Therefore, use of Tramox is not recommended in patients under 12 years of age.

Geriatric patients: In acute pain a dosage adjustment is not necessary as Tramox is given only once or a few times. In chronic pain a dosage adjustment is usually not necessary in elderly patients (up to 75 years) with no clinically manifest hepatic or renal insufficiency. In old patients (above the age of 75 years) elimination may be prolonged. Therefore, if necessary the dosage intervals are to be extended according to the patient's requirements.

Hepatic and renal insufficiency/dialysis: In acute pain a dosage adjustment is not necessary as Tramox is given only once or a few times. In patients with severe renal and/or hepatic insufficiency Tramox should not be administered. In less severe cases prolongation of the dosage interval should be considered.

Duration of Treatment: Tramox must not be given for longer than therapeutically absolutely necessary. If long-term pain treatment is necessary, checks should be carried out at regular and brief intervals (if necessary with breaks in treatment) as to whether and in what doses further treatment with Tramox is necessary.

Route Of Administration

Oral

Contraindication

Tramadol hydrochloride should not be administered to patients who have previously demonstrated hypersensitivity to it or in cases of acute intoxication with alcohol, hypnotics, centrally acting analgesics, opioids or psychotropic drugs. In common with other opioid analgesics it should not be administered to patients who are receiving monamine oxidase inhibitors or within two weeks of their withdrawal.

- Children younger than 18 years to treat pain after surgery to remove the tonsils and/or adenoids.
- Adolescents between 12 and 18 years who are obese or have conditions such as obstructive sleep apnea or severe lung disease, which may increase the risk of serious breathing problems.

Warnings and Precautions

Warnings

At therapeutic doses, Tramadol hydrochloride has the potential to cause withdrawal symptoms. Rarely cases of dependence and abuse have been reported.

At therapeutic doses withdrawal symptoms have been reported at a reporting frequency of 1 in 8,000. Reports of dependence and abuse have been less frequent. Because of this potential, the clinical need for continued analgesic treatment should be reviewed regularly.

In patients with a tendency to drug abuse or dependence, treatment should be for short periods and under strict medical supervision.

Tramadol hydrochloride is not suitable as a substitute in opioid-dependent patients. Although it is an opioid agonist, Tramadol hydrochloride cannot suppress morphine withdrawal symptoms.

Paediatric population

The safety and efficacy of Tramadol has not been studied in the paediatric population. Therefore, use of Tramox is not recommended in patients under 12 years of age.

Respiratory depression

Administer Tramox cautiously in patients at risk for respiratory depression, including patients with substantially decreased respiratory reserve, hypoxia, hypercapnia, or pre-existing respiratory depression, as in these patients, even therapeutic doses of Tramox may decrease respiratory drive to the point of apnea. In these patients, alternative non-opioid analgesics should be considered. When large doses of tramadol are administered with anaesthetic medications or alcohol, respiratory depression may result. Respiratory depression should be treated as an overdose. If naloxone is to be administered, use cautiously because it may precipitate seizures.

Cytochromes P450 (CYP) 2D6 Ultra-Rapid Metabolism

Some individuals may be CYP2D6 ultra-rapid metabolisers. These individuals convert tramadol more rapidly than other people into its more potent opioid metabolites O-desmethyltramadol (M1). This rapid conversion could result in higher than expected opioid-like side effects including life-threatening respiratory depression. The prevalence of this CYP2D6 phenotype varies widely and has been estimated at 0.5 to 1% in Chinese, Japanese and Hispanics, 1 to 10% in Caucasians, 3% in African Americans, and 16-28% in North African, Ethiopians and Arabs. Data are not available for other ethnic groups.

Precautions

Tramadol hydrochloride should be used with caution in patients with head injury, increased intracranial pressure, severe impairment of hepatic and renal function and in patients prone to convulsive disorders or in shock.

Convulsions have been reported at therapeutic doses and the risk may be increased at doses exceeding the usual upper daily dose limit. Patients with a history of epilepsy or those susceptible to seizures should only be treated with tramadol if there are compelling reasons. The risk of convulsions may increase in patients taking tramadol and concomitant medication that can lower the seizure threshold (see 'Interactions').

Care should be taken when treating patients with respiratory depression, or if concomitant CNS depressant drugs are being administered, as the possibility of respiratory depression cannot be excluded in these situations. At therapeutic doses, respiratory depression has infrequently been reported.

Tramadol hydrochloride may cause drowsiness and this effect may be potentiated by alcohol and other CNS depressants. Ambulant patients should be warned not to drive or operate machinery if affected.

1. Risks from Concomitant Use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of Tramox with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking a benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when Tramox is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines (See Drug Interactions).

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AVEX

2. Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of Tramox with serotonergic drugs (See Interactions with Other Medicaments). This may occur within the recommended dosage range. Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal (See Interactions with Other Medicaments). The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue Tramox if serotonin syndrome is suspected.

3. Adrenal Insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness, and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient off of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

4. Sexual Function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility (See Postmarketing Experience)

Interactions

Tramadol hydrochloride should not be administered to patients who are receiving MAO inhibitors or within 2 weeks of their withdrawal.

Concomitant administration of Tramadol hydrochloride with other centrally acting drugs including alcohol may potentiate CNS depressant effects.

Tramadol may increase the potential for both selective serotonin reuptake inhibitors (SSRIs) and tricyclic antidepressants (TCAs) to cause convulsions (see section 'Warnings and Precautions' and 'Pharmacokinetic'). There is a theoretical possibility that tramadol could interact with lithium. There have been no reports of this potential interaction.

Serotonergic drugs: Co-administration with serotonergic drugs, e.g. SSRIs or triptans, may lead to an increase of serotonin associated effects which can include serotonin syndrome.

There have been isolated reports of interaction with coumarin anticoagulants resulting in an increased INR and so care should be taken when commencing treatment with tramadol in patients on anticoagulants.

Pharmacokinetic studies were conducted to investigate the effects of cimetidine, quinidine and carbamazepine on the pharmacokinetics of tramadol.

Carbamazepine

Simultaneous administration of carbamazepine markedly decreases serum concentrations of tramadol to an extent that a decrease in analgesic effectiveness and a shorter duration of action may occur.

Cimetidine

With the concomitant or previous administration of cimetidine clinically relevant interactions are unlikely to occur. Therefore no alteration of the Tramadol hydrochloride dosage regimen is recommended for patients receiving chronic cimetidine therapy.

Quinidine

A study in 12 healthy volunteers has shown that quinidine causes an approximate 25% increase in the tramadol C_{max} and AUC; T_{max} is unaffected. However, the increases in C_{max} and AUC fall within the normal therapeutic range for tramadol, and no dosage adjustment is required.

1. Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ -receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see Warnings and Precautions).

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

2. Serotonergic Drugs

The concomitant use of opioids with other drugs that affect the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue Tramox if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake

inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT₃ receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue) (See Warnings and Precautions).

Pregnancy and Lactation

Pregnancy

Animal studies (rats and rabbits, exposure to tramadol up to 7 times that expected in man) have not revealed teratogenic effects and minimal embryotoxicity (delayed ossification). Fertility, reproductive performance and development of offspring were unaffected. Tramadol has been shown to cross the placenta. There are no adequate and well-controlled studies in pregnant woman. Safe use in pregnancy has not been established. Tramox is not recommended for pregnant women.

Lactation

Tramadol and its metabolites are found in small amounts in human breast milk. Approximately 0.1% of the maternal dose of tramadol is excreted in breast milk. In the immediate post-partum period, for maternal oral daily dosage up to 400 mg, this corresponds to a mean amount of tramadol ingested by breast-fed infants of 3% of the maternal weight-adjusted dosage. For this reason tramadol should not be used during lactation or alternatively, breast feeding should be discontinued during treatment with tramadol. Discontinuation of breast-feeding is generally not necessary following a single dose of tramadol.

Side Effects

Tramadol produces dose-dependent respiratory depression (rare) and sedation of varying degree (from mild fatigue to dizziness), but these symptoms do not generally manifest themselves when moderately severe pain is treated with the recommended oral or rectal doses.

Nausea, sweating, dry mouth, dizziness and giddiness may occasionally occur. Effects on the circulation (palpitations, tachycardia, faintness or circulatory collapse) are possible in rare cases. These adverse effects are particularly liable to occur when in the upright posture, when the drug is given intravenously, or in patients undergoing physical stress.

In addition, headache, nausea, vomiting, constipation, gastrointestinal irritation (abdominal pressure, feeling of fullness) and skin reactions (e.g. pruritis, rashes) may rarely occur. Very rare effects include muscular weakness, alterations in appetite, and disturbances in micturition.

On very rare occasions, tramadol shows a variety of psychological side-effects, varying in nature and intensity from individual to individual, depending on personality and duration of treatment. They include alterations in mood (usually elevated, but sometimes a gloomy mood, known as dysphoria), alterations in activity (usually suppression, sometimes intensification) and alterations in memory (e.g. decision-taking, disturbances of perception). There have been few cases of cerebral convulsions. These however, almost always occurred following intravenous administration of high doses of tramadol with concomitant treatment with neuroleptic agents. Allergic reactions, extending to shock, cannot be reliably ruled out.

Skin diaphoresis has been reported in up to 20% of patients treated with oral/parenteral Tramadol.

Postmarketing Experience:

Serotonin syndrome (See Warnings and Precautions)

Adrenal insufficiency (See Warnings and Precautions)

Androgen deficiency: Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The causal role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility: Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

Symptoms And Treatment of Overdose

Symptoms of overdose are typical of other opioid analgesics, and include miosis, vomiting, cardiovascular collapse, sedation and coma, seizures and respiratory depression.

Supportive measures such as maintaining the patency of the airway and maintaining cardiovascular function should be instituted; naloxone should be used to reverse respiratory depression; fits can be controlled with diazepam.

Tramadol is minimally eliminated from the serum by haemodialysis or haemofiltration. Therefore treatment of acute intoxication with Tramadol hydrochloride with haemodialysis or haemofiltration alone is not suitable for detoxification.

Storage Condition

Store in a dry place protected from light, store below 30°C

Pack size: 10 X 10 blister capsules per box

Manufacturer

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Revised: November 2022